MATH 564

Wednesday August 20

Valve # (k):	1	2	3	4	5	6	7	8
Efficiency E_k:	16.0	32.0	11.7	23.3	14.8	24.5	8.4	53.9
Seating Angle θ_k [°]	3.19	2.11	2.86	3.07	2.95	2.20	2.95	2.62
Clearance d_k [mm]	2.50	2.10	1.61	1.56	2.02	2.13	2.01	2.50
Tension t_k [10^2N]	2.44	2.93	3.18	2.71	2.77	3.04	3.16	3.75

	0.07	4.95	0.48
	2.58	1.35	2.06
	3.72	1.35	1.32
E	1.0	50.7	50.2

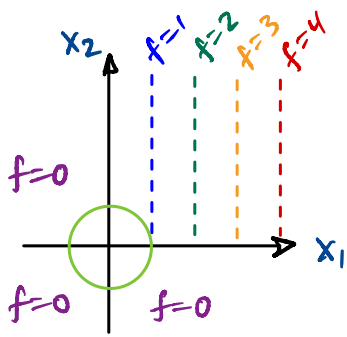
I forgot to mention that there is a canvas page containing this information. Each week update the table.

This is a good start and the results are <u>not</u> unusual. After all, you don't have much information yet to utilize. My expectation is that at some point you will have a dramatic improvement in efficiency followed by slow progress again — unless you are very (un)lucky.

Def 1.3 Function f is said to be continuous at $y \in D$ if for every $\epsilon > 0$, there exists $r > 0$ such that $|f(x) - f(y)| < \epsilon$ for all $x \in N(y,r) \cap D$.

Question: Is the following function continuous at the point $y = (0,0)$?

$$f(x) = \begin{cases} 0 & \text{if } x_1 \le 0 \text{ or } x_2 \le 0 \\ x_1 & \text{otherwise} \end{cases}$$



This function has a "tear" along the positive x-axis.

f is continuous at $y=(0,0)$. Choose $r = \varepsilon$, then for any $\varepsilon > 0$ $|f(x) - f(y)| < \varepsilon = r$ for all x in $N(0,r)$. We see this because $0 \le f(x) < r$ for all x in $N(0,r)$ — see the green circle above.

Derivatives

Def 1.4 The gradient of f, if it exists, is the $n\times 1$ vector $\nabla f(x) = \left[\frac{\partial f}{\partial x_1} \; \frac{\partial f}{\partial x_2} \cdots \frac{\partial f}{\partial x_n}\right]^T \in \mathbb{R}^n$.

Def 1.5 The directional derivative of f in direction $p \in \mathbb{R}^n$, if it exists, is $D_p f(x) = \lim_{a \downarrow 0} \frac{f(x+ap)-f(x)}{a}$. If $\nabla f(x)$ exists then $D_p f(x) = \nabla f^T p = p^T \nabla f$.

Def 1.6 The Hessian of f, if it exists, is the $n\times n$ symmetric matrix

$$\nabla^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

- symmetric
- diagonalizable
- n real eigenvalues

curvature info along eigenvector directions

- basis of eigenvectors

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example $f(x) = \|x\|_2 = (x_1^2 + x_2^2 + \cdots + x_n^2)^{1/2}$

$$\frac{\partial f}{\partial x_k} = \frac{x_k}{\sqrt{x_1^2 + \cdots + x_n^2}} \qquad \text{d.n.e at } x=0$$

$$D_p f(x=0) = \lim_{a \downarrow 0} \frac{\|x+ap\| - \|x\|}{a} \qquad (x=0)$$

$$= \lim_{a \downarrow 0} \frac{|a|\,\|p\|}{a} = \|p\|$$

$$f(x) = \|x\|_2^2 = x_1^2 + x_2^2 + \cdots + x_n^2$$

$$\nabla f(x) = 2x$$

∇f exists everywhere

$$D_p f(x) = \lim_{a \downarrow 0} \frac{\|x+ap\|^2 - \|x\|^2}{a}$$

$$= \lim_{a \downarrow 0} \frac{\sum (x_k + ap_k)^2 - \sum x_k^2}{a}$$

$$= \lim_{a \downarrow 0} \frac{\sum (x_k^2 + 2ap_k x_k + a^2 p_k^2 - x_k^2)}{a}$$

$$= \langle 2x, p \rangle = \langle \nabla f, p \rangle$$

$$= p^T \nabla f$$

as expected when ∇f exists

Def 1.7 Function f is said to be continuously differentiable if ∇f exists and is continuous on D. Furthermore, f is said to be twice continuously differentiable if $\nabla^2 f$ exists and is continuous on D.

Def 1.8 We say that function f is sufficiently smooth (or simply smooth) if any required derivatives of f exist.

In this course we seek to understand optimization on smooth functions. Later in the semester we will extend our skills to address Lipschitz continuous functions.

Extremal Points

Def 1.9 A vector $y \in D$ is a global minimizer (maximizer) of f if $f(y) \le f(x)$ $\left(f(y) \ge f(x) \right)$ for all $x \in D$.

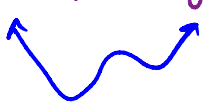
Def 1.10 A vector $y \in D$ is a local minimizer (maximizer) of f if, for some $r > 0$, $f(y) \le f(x)$ $\left(f(y) \ge f(x) \right)$ for all $x \in D \cap \mathcal{N}(y, r)$.

Q: Can a global minimizer be a local minimizer?

Yes, all global minimizers are local minimizers.

Q: Can a local minimizer be a global minimizer?

Yes (sometimes)



2 local mins
1 global min

Q: Can a local minimizer be a local maximizer?

Yes. Ex: $f(x) = a$ ex: $f(x) =$

Q: Is $\nabla f(y) = 0$ at a local minimizer y?

No. We already saw the example $f(x) = \|x\|$ which has minimizer $x = 0$ but $\nabla f(0)$ d.n.e.

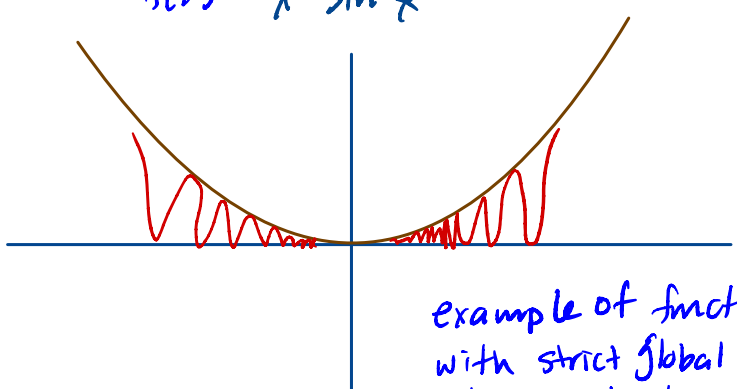
Yes, if f is smooth

Def 1.11 A vector $y \in D$ is a strict global minimizer (maximizer) of f if $f(y) < f(x)$ $(f(y) > f(x))$ for all $x \in D$, $x \neq y$.

Def 1.12 A vector $y \in D$ is a strict local minimizer (maximizer) of f if, for some $r > 0$, $f(y) < f(x)$ $(f(y) > f(x))$ for all $x \in D \cap N(y, r)$, $x \neq y$.

Def 1.13 An extremal point (minimizer or maximizer) y of f is said to be isolated if it is the unique such extremal point in some neighborhood of y.

$f(x) = x^2 \sin^2 \frac{1}{x}$



example of function with strict global min at $x=0$ that is not isolated.